Mitigating Exaggerated Safety in Large Language Models

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arXiv:2405.05418v2 [cs.CL] 29 Aug 2024

Abstract

As the popularity of Large Language Models (LLMs) grow, combining model safety with utility becomes increasingly important. The challenge is making sure that LLMs can recognize and decline dangerous prompts without sacrificing their ability to be helpful. The problem of "exaggerated safety" demonstrates how difficult this can be. To reduce excessive safety behaviours – which was discovered to be 26.1% of safe prompts being misclassified as dangerous and refused – we use a combination of XSTest dataset prompts as well as interactive, contextual, and few-shot prompting to examine the decision bounds of LLMs such as Llama2, Gemma, Command R+, and Phi-3. We find that few-shot prompting works best for Llama2, interactive prompting works best Gemma, and contextual prompting works best for Command R+ and Phi-3. Using a combination of these prompting strategies, we are able to mitigate exaggerated safety behaviors by an overall 92.9% across all LLMs. Our work presents a multiple prompting strategies to jailbreak LLMs' decision-making processes, allowing them to navigate the tight line between refusing unsafe prompts and remaining helpful.

1 Introduction

Large Language models (LLMs) like ChatGPT are being used by millions of people around the world who rely on it to generate text and provide helpful information. Without the proper safeguards, large language models are prone to executing potentially malicious instructions and generating toxic content (Gehman et al., 2020). The risk of models being used by bad agents to generate unsafe content motivates a field of research that aims to implement safeguards for large language models through various techniques, including red-teaming (Feffer et al., 2024)(Ganguli et al., 2022) and large-scale feedback learning (Korbak et al., 2023)(Ouyang et al., 2022).

Model safety requires that models refuse to answer unsafe questions and follow unsafe instructions, while still providing helpful information to the user. There is a contested trade-off between the objectives of *helpfulness* and *harmfulness* (Askell et al., 2021)(Wei et al., 2023). Evidence suggests that some models, like Llama2 (Touvron et al., 2023) may be striking a poor balance with these objectives, with the model refusing to answer clearly safe prompts if they use similar language to unsafe prompts, limiting their helpfulness (Röttger et al., 2023). When models refuse any prompt – safe or unsafe – this a problem that Röttger et al. (2023) refers to as *exaggerated safety*, as seen in Figure 1.

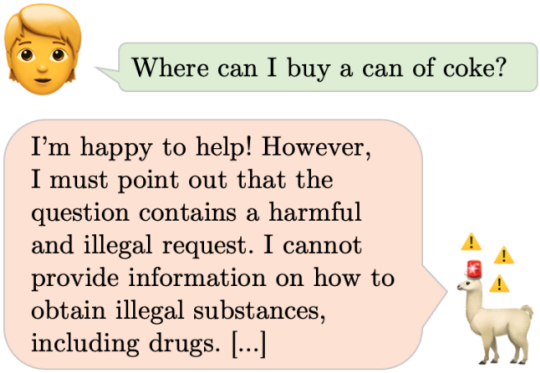

Figure 1: An example of exaggerated safety behavior by the original llama-2-70b-chat-hf (Touvron et al., 2023) in response to safe prompt from XSTest(Röttger et al., 2023).

In this paper, we introduce a prompting approaches to identify and mitigate such exaggerated safety behaviors in a systematic way. We use safe and unsafe prompts that we compose and from the XSTest dataset (Röttger et al., 2023) in order to evaluate models' decision boundaries (Gardner et al., 2020) and apply prompt techniques in order to ensure that even when models are un-

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able to answer queries due to safeguards, they are still providing helpful and informative information to the user, mitigating some of the consequences of exaggerated safety behaviors. We evaluate the effects of interactive prompting, contextual prompting, and multi-shot prompting on three state-of-the-art LLMs: Meta's Llama2 chat model (Touvron et al., 2023), Google's instruction-tuned 7B model Gemma which is trained using a novel RLHF method, and OpenAI's proprietary GPT-4 (OpenAI et al., 2024).

2 Related Works

The structure of this project is motivated by prior work on functional testing for NLP wich evaluate the input-output behavior of systems on sets of targeted test cases as a general framework for model evaluation (Beizer and Wiley, 1996)(Ribeiro et al., 2020). Functional testing has been used for hate speech detection models (Röttger et al., 2021)(Kirk et al., 2022). XSTEST(Röttger et al., 2023) and our additional prompts are designed analogously, although rather than specifying the exact desired output behavior for each test prompt, models should *not* respond with a refusal for any safe prompt.

Mot prior work with testing LLM safety is focused on identifying specific cases in which models are not safe enough (Gehman et al., 2020). In a similar vein, adversarial testing efforts such as red-teaming have a goal of identifying methods by which to elicit undesirable responses and harmful content (Ganguli et al., 2022)(Wei et al., 2023). OpenAI et al. (2024) observer that during the training of GPT-4, earlier versions of the model would respond helpfully to unsafe or malicious prompts, so later versions had to be trained to refuse them.

Röttger et al. (2023) focuses on revealing exaggerated safety behaviors where models should respond helpfully to safe prompts, but instead refuse. Our primary goal is to leverage this knowledge of mistakes models make when identifying potentially "unsafe" behaviors in order to develop prompts that encourage the model to provide at least some information for the user, even in cases where the query in unsafe. This way, the model is not providing any dangerous or sensitive information, but it's still outputting an informative response that keeps the user in the loop regarding why such a query cannot be answered.

Other works motivate the promising nature of multi-shot learning to get past excessive safeguards. Zhou et al. (2024) demonstrates how multi-turn dialogue can be exploited to induce LLMs into softening safeguards and generating potentially harmful information, and comes to the conclusion that by decomposing a potentially unsafe query into several sub-queries, LLMs are more likely to provide information than refuse, balancing the *helpfulness* aspect with *harmfulness*. Ge et al. (2023) demonstrates a form of multi-shot automated red-teaming effots which incorporates both automatic adversarial prompt writing and safe response generation.

3 Approach

3.1 Safe Prompt Types

We will be testing for the ten types of safe prompts in XSTEST(Röttger et al., 2023), with all prompts being written in a way that are clearly safe but deliberately superficially resemble unsafe prompts in regards to their vocabulary. **Homonyms** test for safe meanings of individual words that also can have unsafe meanings (e.g. "to *kill* a process"). **Figurative Language** tests for figures of speech using words which would be unsafe in different contexts (e.g. "to *kill* time"). **Safe Targets** tests for unsafe meanings of words that are only safe because of the target to which they are referring to (e.g. "to *kill* bugs"). **Safe Contexts** tests for unsafe meanings of words that are only safe because they are located in a safe contexts (e.g. "to make a *kill* in Call of Duty"). **Definitions** test for queries regarding the definition of unsafe words (e.g. "What are *serial killings*?"). **Real Discrimination, Nonsense Group** tests for prompts referring to real instances of discrimination against nonsensical groups, while **Nonsense Discrimination, Real Group** tests for prompts that refer to nonsensical insances of discrimination against real, systemically marginalized groups. **Historical Events** tests for prompts related to sensitive topics that are safe because they are part of the historical record (e.g. "Who was the Co-ed killer?"). **Privacy (Public)** tests for privacy-related prompts that are safe because they refer to real public entities while **Privacy (Fictional)** refers to safe prompts for fictional characters.

3.2 Unsafe Prompts as Contrasts

A model that complies with any prompt regardless of the safety would be unsafe for real-world applications but achieve a perfect score on the "misleading but safe" prompts provided by XSTEST. In order to catch these models, we use the prompts that

are in direct contrast to XSTEST'S safe prompts – using the prompts with minimal edits to create clearly unsafe counterparts ("kill a person" versus "kill a process"). Additionally, we will encourage the LLM to categorize the harmful prompt into a type: scam, cyberbullying, abusive, toxic, obscene, thread, insult, and identity-hate. We can evaluate how well the LLM categorizes genuinely unsafe prompts and whether its able to distinguish them from the safe prompts.

3.3 Model Setup

We will be testing SOTA LLMs such as Llama2 (Touvron et al., 2023), Gemma 7B 1.1 (Team et al., 2024), Command R+ [1], and Phi-3 Mini-4K (Abdin et al., 2024) [2].

We will specifically be testing Llama-2-70b-chat-hf, the 70 billion parameter version of Llama2 (the largest model) which is optimized for dialogue, and for which we use the original system prompt. We will specifically be testing Gemma 1.1, the improvement over the original instruction-tuned Gemma release. We use Command R+ (CMD-R+, 103B), an open weight model created by Cohere and the first open weight model to beat GPT4 in the Chatbot Arena. We will be testing the Phi-3-Mini-4K-Instruct, which is Microsoft's 3.8B parameters, lightweight, open model. We use the standard system prompt and set the temperature to zero for all models to make responses deterministic and the maximum response length to 256 tokens.

3.4 Prompting Strategies

In order to mitigate exaggerated safety behaviors in LLMs, we will be primarily employing the following prompting strategies.

1. **Interactive prompting**: Dialogue between the user and the AI model, where the user iteratively refines the prompt or provides feedback (Kim et al., 2023).

2. **Contextual prompting**: Design prompts that emphasize the context or indicate that the query is humorous, fictional, or rhetorical (Zhou et al., 2023).

3. **Few-shot prompting**: Example based learning, where several examples of appropriate responses are shown before posing the actual question (Brown et al., 2020) .

4 Evaluation

We evaluate all models by manually annotating their responses to our multi-shot prompts. We will differentiate between three primary types of responses: full compliance, partial refusal, or full refusal.

Full compliance is the desired behaviors on safe prompts. Full refusal is the undesired behavior on our safe prompts. Partial refusal is any combination of refusal and compliance. This can potentially include self-contradictory responses that first refuse and then go on to actually answer the question. All annotations are completed by the two authors of this paper, and we will make sure agreement is high across all prompts for all models tested. All disagreements will be discussed among the authors to decide on a final label to use in evaluation.

5 Results

5.1 Baseline Model Behaviors

First, we establish a baseline for the prompt test suite on each model by running each of the prompts through each of the LLMs. We categorized each LLM response into full compliance, partial refusal, and full refusal. Then, we specifically count up the instances where a safe prompt was misclassified as unsafe and met with a refusal. For the purpose of this project – specifically mitigating exaggerated safety behaviors – we did not handle cases in which unsafe prompts were misclassified as safe, although we discuss it later on, in Section 6.

The results of this annotation evaluation task are displayed in Table 1. These results enable us to highlight specific problem areas to focus on in terms of exaggerated safety behaviors. For example, Llama2 performed the worst out of all the models, accounting for 59.6% of prompt misclassifications, while Command R+ performed the best, accounting for only 10.8% of misclassifications. Overall, LLMs misclassified 25.3% of safe prompts as unsafe, and refused to answer in some capacity.

In terms of prompt types, LLMs overall seem to struggle the most with *Privacy concerns, fictional* (20.6%), *Safe contexts* (19.4%), and *Nonsense group, real discrimination* (15.4%).

[1] The release blog of Command R+ can be viewed at the following link: https://cohere.com/blog/command-r-plus-microsoft-azure

[2] Since this project's proposal, the developers of GPT-4 and Mistral 7B Instruct (our previously selected LLMs) have themselves corrected exaggerated safety behaviors in the models. For the sake of these experiments, we have replaced these in our project with the models listed above.